INCH-POUND
MIL-DTL-2118D(SH)
28 February 2017
SUPERSEDING
MIL-T-2118C(SH)
22 August 1990

DETAIL SPECIFICATION

TRAP, STEAM, ANGLE, THERMOSTATIC

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers thermostatic stream traps for services up to 100 pounds per square inch (lb/in^2).

1.2 Classification. Steam traps are of the following classes, sizes, and working pressures, as specified (see 6.2).

1.2.1 Classes.

a. Class I - Low pressure (0 to 25 lb/in^2, 268 degrees Fahrenheit [°F] maximum).

b. Class II - High pressure (25 to 100 lb/in^2, 338 °F maximum).

1.2.2 Sizes. Sizes are listed in table I and range from ½ to 2 inches nominal pipe size (nps) on the inlet side of the trap.

1.2.3 Working pressures. Class I includes working pressures of 5 lb/in^2, 15 lb/in^2, and 25 lb/in^2. Class II includes working pressures of 25 lb/in^2, 35 lb/in^2, 50 lb/in^2, and 100 lb/in^2.

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line “Document Comment”. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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1.3 Part or identifying number (PIN) (see 6.2).

1.3.1 Steam trap assembly. The PIN for steam trap assemblies should be constructed as follows:

M	**2118**	-	**X**	-	**X**	-	**X**	-	**X**
Prefix for Military Specification	Specification Number		Class and working pressure (see table I)		Size (see table I)		S – Standard configuration or N – Nonstandard configuration (see 3.4.3)		P – Nickel plated (see 3.8) or O – When not nickel plated

Example: A Class II, 25 lb/in² working pressure, Size 5, Standard configuration, Nickel plated steam trap would be PIN M2118-D-5-S-P.

1.3.2 Steam trap thermostatic element. The PIN for steam trap thermostatic elements should be constructed the same as for the steam trap assembly, except configuration and plating will not apply; for example, M2118-D-5.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.2).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-1183 - Fittings, Pipe, Cast Bronze, Silver-Brazing, General Specification for

(Copies of this document are available online at http://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Copies of this document are available online at http://webstore.ansi.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.4.

3.2 Material. The material for each part shall be as specified herein. However, where a definite material is not specified, an appropriate ASTM, AISI, or federal (global) specification material shall be used which will enable the traps to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guarantee of the acceptance of the finished product (see 4.5.1).

3.2.1 Covers and bodies. Covers and bodies shall be made of ASTM, AISI, or Federal Specification brass, bronze, or copper-nickel alloy acceptable for steam service.

3.2.2 Valve pins and renewable seats. Valve pins and renewable seats shall be made of the following ASTM, AISI, or Federal specification materials: corrosion-resisting (chrome) steel, nickel, or copper-nickel alloy.

3.2.3 Thermostatic elements. Thermostatic elements shall be of corrosion-resisting materials capable of long life without physical changes, such as cracking, softening, or stretching, at the maximum pressure and temperature required for the class.

3.3 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.4 Construction. Thermostatic elements and valve pins of the same design and manufacture for any given size of a class shall be interchangeable, without adjustment, in that size and class. Thermosensitive valves, when placed in the trap bodies and with covers properly secured, shall be correctly positioned and distanced from the valve seats (see 4.5.1).

3.4.1 Thermostatic elements (and strainers when provided). Traps shall be easily removed for examination and cleaning without breaking the inlet and outlet connections. Removal of the element while hot shall not result in distortion (of the element) that would affect future operation.

3.4.2 End connections. End connections of traps shall be union ends with silver-brazed tailpieces in accordance with MIL-DTL-1183.

3.4.3 Trap configuration. Unless otherwise specified (see 6.2), traps shall be of the angle pattern with side inlet and bottom outlet.

3.4.4 Class I traps. Strainers, as specified for Class II traps, may be provided at the option of the manufacturer.

3.4.5 Class II traps. Each Class II steam trap shall be provided with a rugged strainer of ASTM, AISI, or Federal Specification corrosion-resisting material. The strainer shall be integral with the trap inlet side, if possible, but it may be furnished in a separate fitting connected to the trap on the inlet side.

3.4.6 Valve joints. Joints in the thermostatic valve element shall be mechanical, soldered, or a combination of both. Where solder is used for Class I traps, it shall have a softening point of not less than 315 °F. Where solder is used for Class II traps, it shall have a softening point of not less than 425 °F.

3.4.7 Valve seats. Class II traps shall have renewable valve seats.

3.5 Performance.

3.5.1 Steam leakage. When exposed to saturated steam at working pressure, the thermostatic element shall cause the steam trap valve to close, causing a major reduction in flow through the steam trap (see 4.5.2).

3.5.2 Discharge capacity. Steam traps shall discharge the minimum continuous discharge capacities specified in table I (see 4.5.2) for the size, class, and working pressure specified.

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TABLE I. Minimum continuous discharge capacity (pounds of water per hour).

Size	Trap inlet nps (inches)	Class and working pressure (lb/in²)						
		Class I			Class II			
		A (5)	B (15)	C (25)	D (25)	E (35)	F (50)	G (100)
1	½	500	900	1,200	1,200	1,550	1,950	2,600
2	¾	900	1,400	1,800	1,800	2,200	2,600	3,500
3	1	1,300	2,000	2,500	2,500	2,900	3,400	4,500
4	1¼	3,000	5,000	6,500	6,500	7,500	8,500	11,000
5	1½	4,000	7,000	9,000	9,000	10,000	12,000	18,000
6	2	5,000	9,000	11,000	11,000	14,000	16,000	21,000

3.5.3 Temperature differentials. Class I and Class II traps in table I shall function within the following temperature differentials (the difference between the temperature of the discharge water and that of the saturated steam of the same pressure as on the trap). Class I traps shall start discharging with a temperature differential of 10 to 25 °F. Class II traps shall start to discharge at a temperature differential of 30 to 40 °F (see 4.5.2).

3.5.4 Hydrostatic strength. Thermostatic traps shall be hydrostatically tested to 150 percent of the maximum working pressure for the class of service to determine that trap bodies and covers are not porous. Class I traps shall be tested to a hydro (body) cold rating pressure of 37.5 lb/in², and Class II traps shall be tested to a hydro (body) cold rating pressure of 150 lb/in² (see 4.5.3). Hydrostatic testing of Class I and Class II traps shall be conducted without thermostatic valves installed.

3.5.4.1 Pressure rating. Thermostatic traps shall have a maximum allowable working pressure of 25 lb/in² for Class I and 100 lb/in² for Class II.

3.6 Identification marking. Each trap shall be marked legibly and permanently as specified below:

a. Manufacturer’s name or trademark (see 3.6.1).

b. Model number (see 3.6.1).

c. Pressure class number (for example, CL. II – H.P.).

3.6.1 Manufacturer’s name or trademark and model designation. The manufacturer’s name or trademark and model designation shall be permanently and visibly marked on a non-operating part.

3.7 Special tools. Steam traps shall not require special tools for installation and routine maintenance.

3.8 Exterior finish. Machined surfaces shall not exceed 125 roughness height rating (RHR) as defined in ANSI B46.1. When specified (see 6.2), castings shall be nickel-plated internally and externally.

3.9 Workmanship. Components shall be free from defects which affect their appearance or which may affect their operation (see 4.5.1). Castings shall be clean, sound, smoothly cored, true to form, uniform in texture, and free from cold shunts, porosity, blow holes, and other injurious defects. Castings shall be thoroughly cleaned, both inside and outside, and all fins and roughness shall be removed.

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified (see 6.2), inspections shall be performed in accordance with the conditions specified in 4.2.1 through 4.2.2.

4.2.1 Test temperature. Inspection and tests shall be accomplished at ambient room temperature.

4.2.2 Test fluid. Tests shall be conducted using saturated steam at the pressures identified in 3.5.4.

4.3 First article inspection. First article inspection shall be performed on the thermostatic element design when a first article sample is required (see 3.1). This inspection shall include the examination of 4.5.1 and the tests of 4.5.2 and 4.5.3. The element may be installed in either a simulated steam trap or an actual steam trap as long as the chamber dimensions where the element is mounted are equal. Inspection shall consist of the performance test of 4.5.2 and shall be caused to cycle at least ten times. A visual inspection of the element after this test shall show no signs of cracking, deformation, weakness of the element, or erosion of the seating element.

4.4 Conformance inspection. Conformance inspection shall be performed on sample units produced with equipment and procedures normally used in production.

4.4.1 Lot. Traps of the same material, type, style, size, and class offered for delivery at the same time are considered to be a lot.

4.4.2 Sampling. A random sample shall be selected from each lot for conformance inspection. The sample shall be in accordance with table II.

TABLE II. Sampling for examination and tests.

Lot size, number of traps	Samples, number of traps to be examined and tested
2 to 15	7
16 to 25	10
26 to 65	15
66 and over	20

4.4.3 Inspection. The sample items selected in accordance with 4.4.2 shall be subjected to the conformance inspection shown in table III.

TABLE III. Conformance inspection.

Inspection	Requirement	Test method	Notes
Visual and dimensional examination	3.1 through 3.4.7, 3.6, 3.8 and 3.9	4.5.1	
Performance test	3.5.2 and 3.5.3	4.5.2	First article only
Hydrostatic strength test	3.5.4	4.5.3	Housing only

4.4.4 Order of test. Tests shall be conducted in the order shown in table III.

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4.4.5 Disposition of sample units. When specified (see 6.2), sample units which have been subjected to conformance tests and approved may be delivered on the contract or purchase order in an assembled, ready-for-service condition.

4.5 Methods of inspection.

4.5.1 Visual and dimensional examination. Each sample selected in accordance with 4.4.2 shall be visually and dimensionally examined to determine conformance with this specification, including, but not limited to, the requirements of 3.6, 3.8, and 3.9. Major and minor defects shall be classified as shown in table IV. Any sample failing to conform to the visual and dimensional examination shall be cause for rejection of the entire lot.

TABLE IV. Classification of defects.

Major	
101	Class, pressure, and size not as specified.
102	Improper assembly or valve pins not interchangeable.
103	Improper materials used.
104	Dimensions incorrect or out of tolerance
Minor	
201	Identification markings missing, incorrect, or illegible.
202	Color code painting inadequate.
203	Workmanship requirements not met.

4.5.2 Performance test. The first article elements shall be operated at the conditions specified in table I and 3.5.3 to determine that the discharge capacity conforms to 3.5.2. When subjected to the saturated steam temperature and pressure, the element shall close and conform to 3.5.1.

4.5.3 Hydrostatic strength test. All steam trap housings shall be tested to comply with the requirements of 3.5.4. Test pressure shall be held for 3 to 5 minutes. The unit shall not rupture (sustain a fracture of the body or cover) or become deformed.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Thermostatic steam traps built to this specification are used to drain low-pressure (Class I) or high-pressure (Class II) steam lines aboard naval ships; can be maintained on deployment by shipboard personnel; and are designed to last for the life of the ship. Existing stock of thermostatic steam traps manufactured to MIL-DTL-2118C(SH) dated 22 August 1990 are acceptable for use until depleted.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2).

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c. Applicable part or identifying number (PIN) (See 1.3).

d. When first article is required (see 3.1).

e. Trap configuration when nonstandard is specified (see 3.4.3).

f. When castings are to be nickel-plated internally and externally (see 3.8).

g. Inspection conditions, if other than specified (see 4.2).

h. Disposition of sample units (see 4.4.5).

i. Packaging requirements (see 5.1).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should allow the Government to reserve the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Subject term (key word) listing.

Equipment, galley

Equipment, laundry

Heating coil

Steam radiator

Temperature, differential

Thermostatic element

Thermoventilating heating unit

Union end

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4730-2017-034)

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